Problem

A company wants to invest an amount of money in n financial assets (share, bonds, etc.). It wants to decide which fraction to invest in each asset $i = 1, \ldots, n$. Let x_i be the fraction invested in asset i. Acquiring assets has expenses of $f_i\%$ for each asset i. The future returns are given by the random variables ξ_i for each asset, which are assumed to be independent. Formulate the two-stage stochastic optimization problem to be solved by the company. Simplify as much as possible the second stage, obtaining the deterministic equivalent formulation.

Solution

- $x_i, i = 1, \ldots, n$: fraction invested in asset i.
- $y_i, i = 1, \ldots, n$: benefit from asset i.

$$
\begin{array}{ll}
\min & \displaystyle\sum_{i=1}^{n} f_i x_i + E[Q(x,\xi)] \\
\text{s.to} & \displaystyle\sum_{i=1}^{n} x_i = 1 \\
& 0 \leq x_i \leq 1 \quad i = 1, \ldots, n
\end{array}
$$

where

$$
Q(x,\xi) = \begin{array}{ll}
\min & -\displaystyle\sum_{i=1}^{n} y_i \\
\text{s.to} & y_i = \xi_i x_i
\end{array} = -\sum_{i=1}^{n} \xi_i x_i.
$$

The deterministic equivalent formulation is thus

$$
\begin{array}{ll}
\min & \displaystyle\sum_{i=1}^{n} f_i x_i - E[\sum_{i=1}^{n} \xi_i x_i] = \sum_{i=1}^{n} (f_i - E[\xi_i]) x_i \\
\text{s.to} & \displaystyle\sum_{i=1}^{n} x_i = 1 \\
& 0 \leq x_i \leq 1 \quad i = 1, \ldots, n
\end{array}
$$

In this problems there is no recourse action. The second stage variables are only auxiliary variables to compute the future expected benefits.